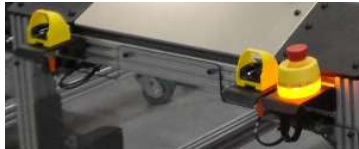


Electrical Engineering Line-up

ASI Job #: 8802
Qty. (1)
Customer: Performance Seats, Spartanburg, SC
Package Release: 09/11/26
Hardware Due: 09/14/26
Panel Complete: 09/16/26
Runoff Date: 09/25/14
Fixture Description: Blackbird 1st Row FSC Build Pallet with Transfer Cart and Docking Station
PLC Enclosure: 24" tall x 30" wide x 8" deep ~ Electro-Matic Pt. #EMRA3WM8017549A015
HMI Enclosure: 20" tall x 20" wide x 8" deep
Support Arm Kit: N/A (Tekno swivel as described)
PLC: Allen-Bradley Compact Logics – Part #1769-L24ER-QB1B
HMI: Maple Systems 15" Smart Touchscreen – Part #CMT2158XV2
PLC IP: xxx.xxx.x.xx
TC #1 IP: Subnet Mask: 255.255.00
Gateway:

Cycle Start Style:



Reference Job: 8688 (base hardware, cycle start)

Fixture Outline:

Automated Solutions will design and manufacture (1) FSC Carbon Fiber Tooling Cart. Tooling Cart are to be secured in, Performance Seats, existing fixture (8466). ASI Overhead Torque Arm package to be used. Maple Systems 15" HMI, and Atlas Copco Torque package are recommended, and shown as option below. Cost for this option include integration onto existing fixture (8466). ASI assumes existing fixture has the capability to support the above-mentioned additions, both structurally and electrically. ASI is not responsible for any additional cost, should additional components be needed to support the above mentioned. Performance Seats must provide latest part data, in STEP format, before start of project. Performance Seats will ship existing fixture to ASI at start of project. ASI will follow a standard Automated Solutions machine and controls package built to Performance Seats Plant specifications.

Machine Home Condition:

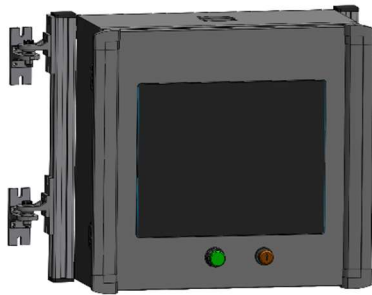
1. Clamp cylinder(s) retracted
2. Part present sensor(s) OFF

Anticipated Sequence of Operations:

1. Operator loads seat pan into fixture.
2. Driver / Passenger is determined via cube sensors.
3. Operator, vis Opto Touch, cycles fixture.
4. Seat cushion is clamped.
5. Operator changes to the appropriate bit.
6. Operator, via 90-degree torque gun, removes (X2) seat pan side bolts.
7. Operator, via screw gun, removes (X4) seat pan screws.
8. Operator discards steel seat pan from cushion frame.
9. Operator manually sets carbon fiber seat pan into position.
10. Operator, via 90-degree torque gun, secures (X2) seat pan side bolts first.
 - a. Torque value: 15 +/- / 1.5Nm
11. Operator, via screw gun, secures (X4) seat pan top screws.
12. Operator, via Opto touch, cycles fixture and seat cushion frame is unclamped.
13. Cycle complete.

Major components on machine:

- 120VAC main power with 24VDC control voltage – include standard rotary disconnect
- (x1) Overhead LED light required ~ Banner Pt. #WLB32ZC285MQMB
 - Tie to timer relay but also include button on HMI
- (x1) Metal Power Strip ~ Amazon Pt. #B0B54N4LVQX
- (x1) Allen-Bradley Compact Logics PLC ~ Pt. #1769-L24ER-QB1B
- (x1) HMI Assembly includes all below
 - (x1) Maple Systems 15" Smart Touchscreen ~ Part #CMT2158XV2
 - (x1) Enclosure ~ Rittal Pt. #1060250 (includes sub-plate)
 - (x4) 1x1 Extrusion ~ Tekno Pt. #E0100201 x 19.50" Lg. (MBOM)
 - (x4) 1x1 End Cap ~ Tekno Pt. #H0400221 (MBOM)
 - (x2) Swivel Adapter ~ Tekno Pt. #B070051100 (MBOM)
 - (x4) End Plate ~ Tekno Pt. #B0300121 (MBOM)



- (x1) Allen-Bradley Green System Start Button ~ Pt. #800T-QT24G
 - Mounted inside HMI box (see image above)
- (x1) Allen-Bradley Auto/Manual Key Switch ~ Pt. #800T-H42A
 - Mounted inside HMI box (see image above)
- (x1) Dual Ethernet programming port ~ Mencom Part # MEN-DP1-2RJ45-16LS (Use Existing)
 - Mount on side of enclosure
- (x1) Stack Light ~ Banner Pt. #TL50GRBAQ (Use Existing)
 - (x1) Cable for stack light ~ IFM Pt. #EVC071(Replace Existing)
- (x1) E-stop Button ~ Banner Part #SSA-EB1PLYR-12ECQ8 (Use Existing)
 - (x1) IFM QD Cable for E-stop ~ Part #E11807 (Use Existing)
- (x1) E-Stop Relay ~ Banner Part #ES-FA-9AA (Use Existing)
- (x2) Banner Opto Touch Button ~ Part #STBVR81Q6 (64192) (One Included on System)
 - (x2) IFM QD Cable for opto touch ~ Part #EVC071(One Included on System)
- (x1) Cycle Start Relay ~ Banner Part #AT-FM-10K (60698) (Use Existing)
- (x1) M12 4-Pin Male to RJ45 5M Ethernet Cable ~ Balluff Pt. #BCC0E3A
- (x1) M12 5-Pin Female L-Code to Flying Lead 5M Power Cable ~ Balluff Pt. #BCC0L2P

- (x1) Atlas Copco Torque Tool System (ASI to Supply)
 - (x1) Power Focus 8 Torque controller ~ Pt. #8436280002 (put on EBOM)
 - (x1) IAM PF 32GB Critical 1VS Storage Unit ~ Pt. #8436095009 (put on EBOM)
 - (x1) PF Tool Cable 10M ST ~ Pt. # 4220437510 (put on EBOM)
 - (x1) 90° ETV-STR61-20-10 gun ~ Pt. #8436613020 (put on EBOM)
 - (x1) Anybus CC – EtherNet/IP AC40 ~ Pt. # 8436094115 (put on EBOM)
 - Standard bad torque & re-try key switch logic not required
 - 1st attempt is primary / 2nd, 3rd & 4th attempts are re-hit
 - Lockout after 2 total attempts / Reset counter once good torque achieved
 - (x11) P-sets required
 - (x1) LH Seat Pan Bolt
 - (x1) RH Seat Pan Bolt
 - (x1) OCS Bracket Bolt
 - (x4) Driver Cushion Pan Screws (May not need)
 - (x4) Passenger Cushion Pan Screws (May not need)
 - Torque value
 - Seat Pan Bolt: 15 +/-1.5 Nm
 - OCS Bracket Bolt: 1.8 +/-0.2Nm
 - Torque sequence required
 - LH, RH

Mechanical components for reference (on MBOM):

- (x1) Custom aluminum extrusion frame
- (x4) Linak Ergo jacks & controller
- (x1) 3/8" Air Prep ASI. standard ~ SMC Pt. #FG05024
- (x1) Manifold Sub-Assembly required includes all below
 - (x1) 4-Station manifold ~ SMC Pt. #VV5QC41-04N7TFD0
 - (x1) 3 Position Normally open valve. # VQC4301-51 **VALVE #1**
 - (2) Clamp Cylinder – Station #1
 - (x3) Blank plate # VVQ4000-10A-1 **VALVES #2, #3 & #4**
- (x2) Part present sensor for LH/RH cushion frame
- (x3) Balluff I/O Linear Transducers for reaction arm position on X, Y, and Z-Axis
- (x1) I/O-Link Master block ~ Pt. #BNI00K6

Inputs:

- (1) Control Power On
- (1) Auto Mode Enable Key Switch
- (1) Cycle Start Relay
- (2) Part Present Sensors
- (2) Position sensor for reaction arm position on X-Axis
- (2) Adv/Ret Clamp Cylinder Switch
- (1) E-Stop Relay

10 Inputs

Outputs:

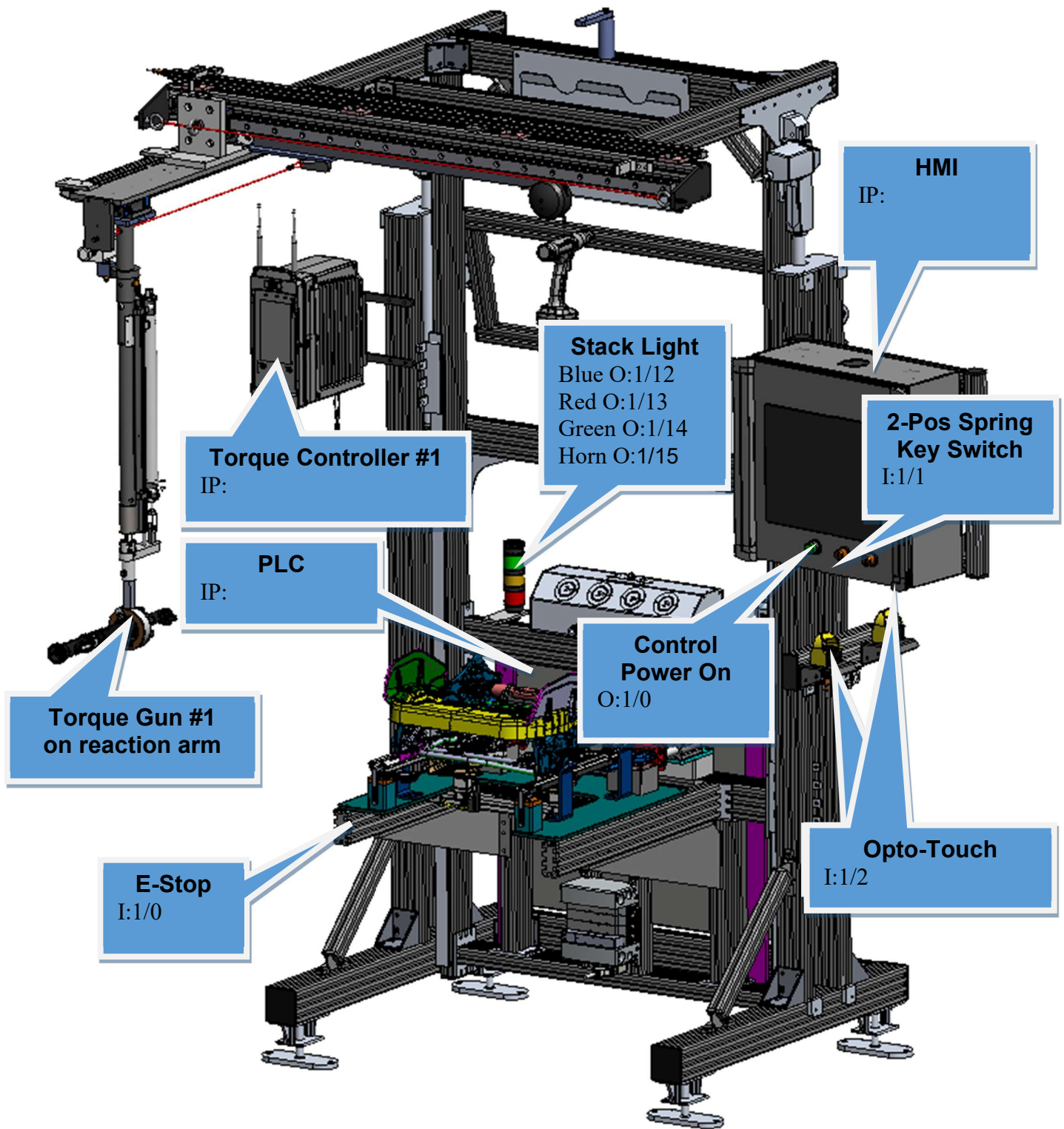
- (1) Control Power on Button Light
- (2) Adv/Ret Clamp Cylinder
- (6) Adv/Ret Spare Valves
- (4) Stack Light
- (1) Overhead Light Relay

14 Outputs

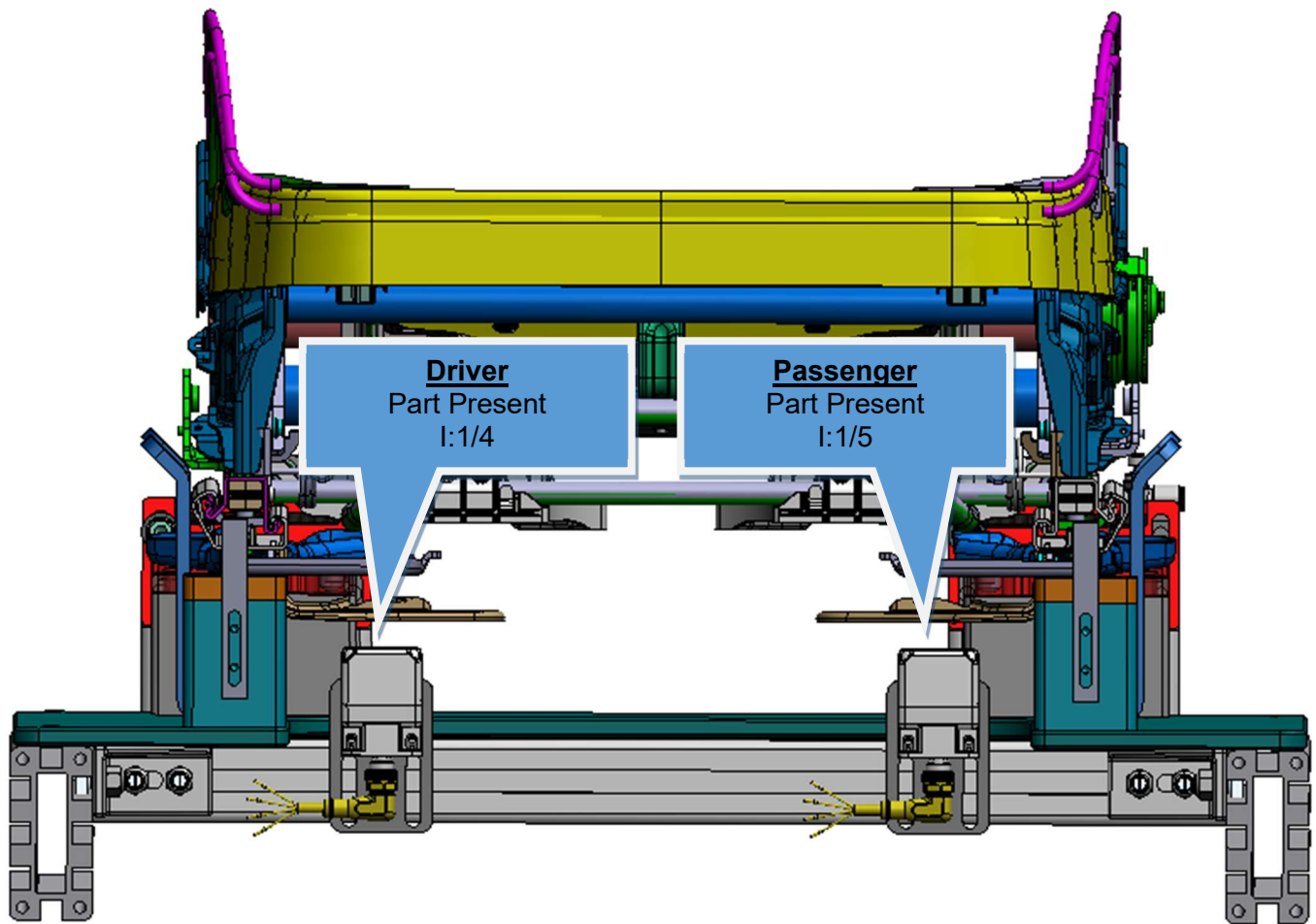
Station Shown with Notes/Comments:

- ASI standard hardware and controls practices
- LED light board not required
- Please ensure 30% spare I/O
- Leave valves energized in their work and home states
- Lockout stickers will be required P1, E1
- Ensure all PP are labeled with I/O address

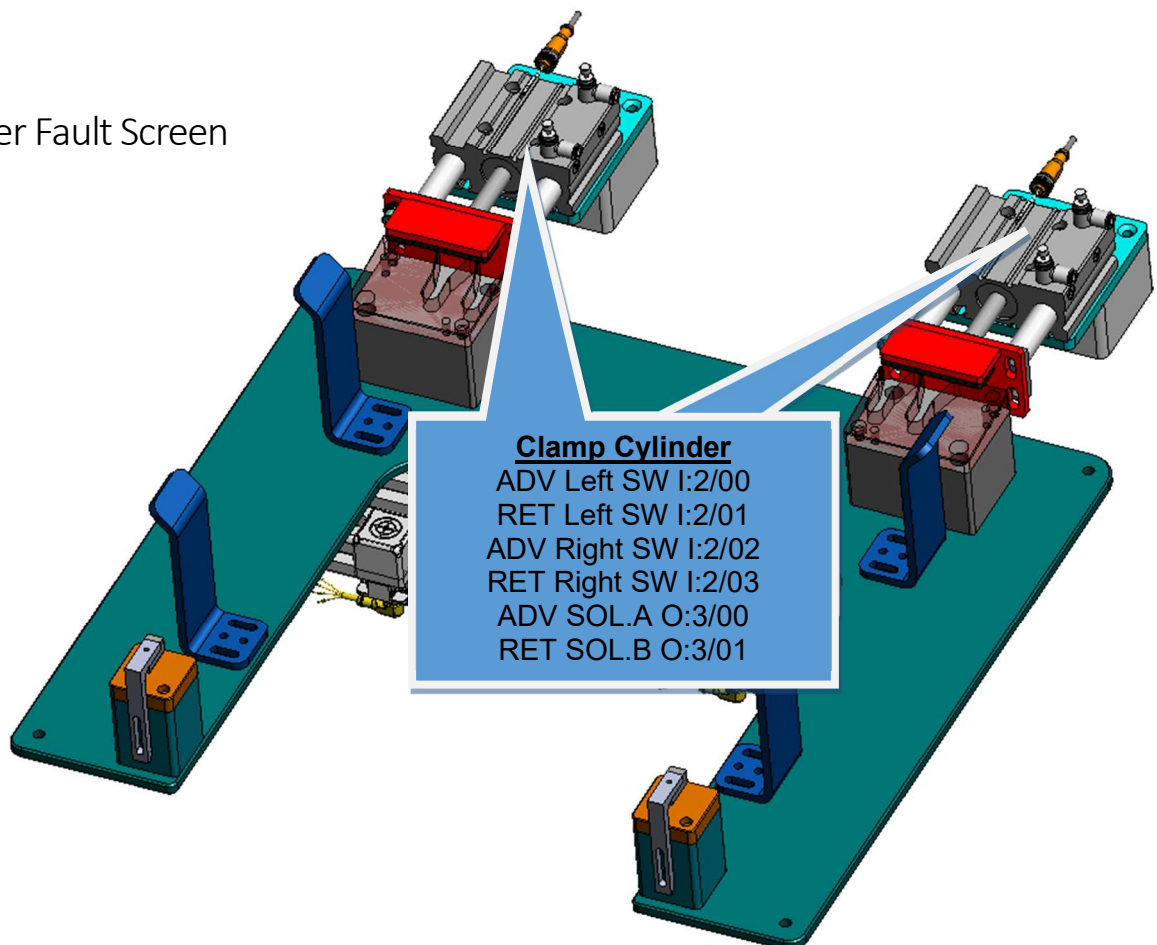
Station Layout



Tooling Layout Screen



Cylinder Fault Screen



Torque Screen

